Week # 12: 16–20 November 2026

Subject / الموضوع: Computer Science **Unit Title / عنوان الوحدة: Cybersecurity Principles and Threats** **Grade 11 / الصف الحادي عشر**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1 (W12.1) الحصة الأولى **Patterns in Digital Data CS.HS.4.2(g, i) 55 min**	I can explain what digital records reveal about daily routines by finding two patterns in a fictional travel log.	Formative draft · Start the shared Digital Patterns and Encryption case study. Check · Which entries support each pattern? 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 6.2 Support planning with templates, strategy prompts, and modeled thinking.	Fictional Abu Dhabi travel log, encryption diagram and class case-study sheet. At home, discuss what repeated timestamps can reveal about a routine.
Lesson 2 (W12.2) الحصة الثانية **Brute Force and Encryption Choices CS.HS.4.2(h, j) 55 min**	I can explain brute force by counting possible codes and linking the model to encryption keys.	Formative · Complete and submit the same case study for feedback. Check · Why does a larger set of possible codes take longer to try? 5.3 Build fluency with graduated scaffolds and models. 8.4 Provide mastery-oriented feedback that emphasizes growth and improvement.	Two- and three-digit code cards, optional calculator and the same case-study sheet. At home, explain why a longer code gives more combinations.
Homework الواجبات	Complete the case study in class.		Keep your Lesson 1 draft ready for Lesson 2.
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Use the same fictional travel log in both lessons. Compare 00–99 and 000–999, with leading zeros and no lockout. Link the model to testing every possible encryption key. Show when encryption protects a file and how a weak passphrase or careless sharing can still expose the data.		

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Technology Integration دمج التكنولوجيا	Display a fictional travel log and a diagram of encryption during storage and transfer. Use paper code cards or a calculator to model exhaustive guessing.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics — compare 10^2 and 10^3 possible codes. Moral Education and UAE Studies — discuss responsible handling of travel records and respect for people's privacy.
External Assessment Integration دمج التقييمات الخارجية	One classroom formative · Digital Patterns and Encryption case study. Students start the draft in Lesson 1 and submit the completed case study in Lesson 2.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Critical thinking — separate patterns from assumptions and weigh a security trade-off. Communication — explain a claim using evidence. Offer a visual model, word bank and sentence starters.
Standards المعايير	Colorado Computer Science (2024) · High School Standard 4 · CS.HS.4.2 · PG 6, recognise and analyse security concepts. Lesson 1 (g, i) · Describe how everyday systems reveal routines and how encryption is used. Lesson 2 (h, j) · Explain exhaustive guessing and evaluate an encryption method in context.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	What can digital records reveal about a person's routine, and how can we protect those records while allowing authorised use?
Success Criteria معايير النجاح **(Measurable)**	Lesson 1 · MUST identify two patterns and describe uses of encryption in storage and during transfer. SHOULD cite entries that support each pattern. COULD describe one risk that remains after encryption. Lesson 2 · MUST calculate the number of codes, explain why exhaustive guessing reaches the right key given enough time, and identify one strength and one weakness of password-based file encryption. SHOULD explain why both matter for the log. COULD justify an extra safeguard.
National Identity Domain مجالات الربط بالهوية الوطنية	**Citizenship** Belonging الإنتماء | **Values** Respect الاحترام | **Culture** Arabic Language اللغة العربية
	Area / المجال Responsible digital citizenship and privacy in UAE digital services.
	Implementation / التطبيق Explain why a fictional Abu Dhabi transport service should protect journey records. Use clear English and Arabic terms when discussing privacy.